

Introduction to neural networks

Parametric models

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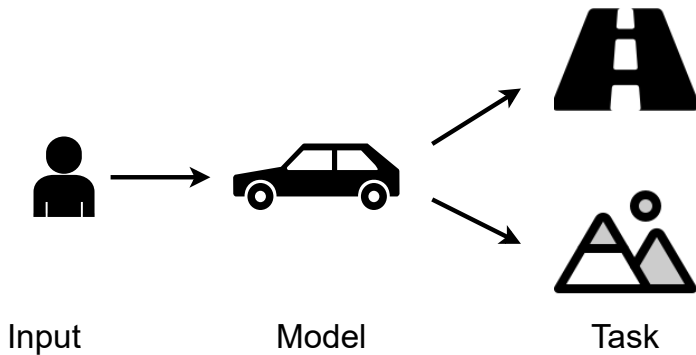
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- 1 Parametric models
- 2 Nonparametric models

Parametric models



Parametric models

Features

Parametric model

A learning model that summarizes data with a set of parameters of fixed size is called a parametric model [Russell and Norvig].

- Reduces the data spatial complexity
- Standardizes the training algorithms
- To define the model size is a new problem

Parametric models

General model:

$$\hat{Y} = \Phi_{\theta}(X) \quad (1)$$

Example

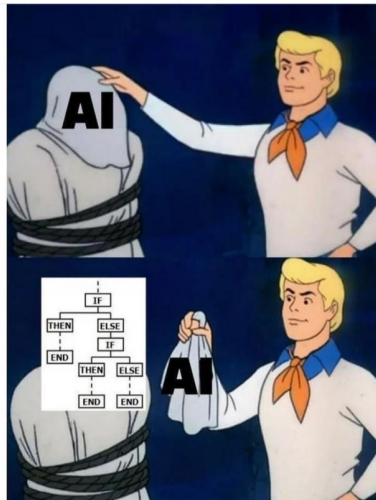
$$y = mx + b \quad (2)$$

$$\theta = ? \quad (3)$$

- A nonparametric model is one that can not be characterized by a bounded set of parameters.
 - E.g. instance based learning
 - Nearest neighbors
 - Nonparametric regression

Nonparametric models

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- RUSSELL, Stuart J.; NORVIG, Peter. Inteligencia Artificial: un enfoque moderno. Capítulo 18 Learning from examples. 2004.
- Photo by Tommy Lisbin on Unsplash
- Photo by Hanson Lu on Unsplash